

Muhammad Ali Waris Khan

AI / RAG Engineer — Generative AI Developer

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Professional Summary

AI/RAG Engineer with hands-on experience building production-style document intelligence systems using Python, LangChain, Gemini AI, and FAISS. Experienced in developing end-to-end pipelines for document extraction, enrichment, chunking, embedding generation, indexing, and semantic retrieval. Comfortable building modular backend services, debugging production workflows, and developing scalable AI applications, with working knowledge of Azure AI architecture and cloud-native document intelligence systems

Technical Skills

- **Programming:** Python, AsyncIO (Async/Await), Threading, Concurrent Programming
- **Backend Engineering:** Python Services, REST APIs, HTTP, JSON, Git, Modular Architecture, Exception Handling
- **LLM Engineering:** Gemini AI, LangChain, Prompt Engineering, Conversational AI, RAG Pipelines, Hallucination Mitigation
- **Document Intelligence:** PDF Parsing (PyMuPDF), Chunking, Enrichment, Vectorization, Indexing
- **Retrieval & Vector Search:** FAISS, Semantic Search, Dense Embeddings, MMR Retrieval, Cross-Encoder Reranking
- **Machine Learning / NLP:** Sentence Transformers, Hugging Face Embeddings, NLP Preprocessing
- **Frontend & Deployment:** Streamlit, Streamlit Cloud; Familiar with Azure cloud concepts including Blob Storage, Azure AI Search, Function Apps, and Azure OpenAI architecture.
- **Software Engineering:** Debugging, Stateful Session Management, Logging, Reproducible Pipelines, Production-Oriented Development
- **Development Tools:** Git, GitHub, VS Code, Cursor AI
- **Libraries & Tools:** PyMuPDF, yt-dlp, youtube-transcript-api, sentence-transformers, dotenv

Deployed Projects

1. Production-Style Document Intelligence Platform

(Python, LangChain, Gemini AI, FAISS, Streamlit, PyMuPDF)

- Engineered and deployed a production-style document intelligence platform implementing document extraction, enrichment, chunking, embedding generation, vectorization, indexing, and semantic retrieval for multi-PDF interaction.
- Built a multi-stage retrieval pipeline using Gemini embeddings, FAISS, MMR retrieval, and Cross-Encoder reranking to improve search relevance and grounded responses.
- Developed modular Python services for document ingestion, indexing, retrieval, conversational memory, and response generation following reusable backend architecture principles.
- Implemented streaming responses, page-level citations, source traceability, and conversational memory for explainable multi-turn question answering.
- Improved pipeline reliability through validation, exception handling, and logging across document ingestion and retrieval workflows.

- Deployed the application on Streamlit Cloud. Live Demo: pdfcantalk.streamlit.app

2. AI Resume Analyzer with ATS Scoring

(Python, Gemini AI, Streamlit, PyMuPDF, Regex NLP)

- Developed AI-powered ATS resume analyzer evaluating resumes against job descriptions through NLP keyword analysis and skill matching
- Built robust automated PDF parsing and preprocessing pipeline with validation and exception handling
- Engineered custom ATS scoring algorithm with missing-skill detection and compatibility scoring
- Integrated Gemini AI for contextual optimization feedback and recruiter-style recommendations
- Implemented explainable interface displaying matched/missing keywords and improvement suggestions
- Designed modular backend architecture with reusable Python components and secure API integration.
- Deployed on Streamlit Cloud. Live Demo: atsresume-analyzer.streamlit.app

3. Transcript Intelligence RAG Platform

(Python, Gemini AI, LangChain, FAISS, Sentence Transformers, Streamlit)

- Engineered a transcript intelligence platform for semantic interaction with YouTube video transcripts.
- Implemented transcript intelligence pipeline with metadata extraction, timestamp-aware chunking, and semantic indexing
- Engineered multi-stage retrieval combining FAISS, MMR, and Cross-Encoder reranking for high precision
- Integrated conversational memory and streaming LLM responses for real-time multi-turn interaction
- Developed timestamp-based citation system linking answers directly to original video segments
- Implemented persistent vector storage with incremental indexing for scalable workflows
- Deployed on Streamlit Cloud. Live Demo: chatwithvideos.streamlit.app

4. Academic Document Intelligence Platform

(Python, Gemini AI, LangChain, FAISS, Hugging Face, Streamlit)

- Developed an academic document intelligence platform for semantic search, literature review generation, and research paper analysis.
- Built multi-document RAG pipeline with Hugging Face embeddings and FAISS for semantic retrieval
- Implemented multi-stage retrieval with reranking for improved precision and grounded responses
- Engineered automated summarization, methodology extraction, and literature synthesis using Gemini AI
- Applied hallucination-aware prompt engineering for context-grounded academic question answering
- Built modular production architecture with explainable citations and robust validation
- Deployed on Streamlit Cloud. Live Demo: chatwithresearchpaper.streamlit.app

Education

Masters in Artificial Intelligence

College of Electrical and Mechanical Engineering, NUST, Islamabad

Links

- github.com/maw-khan
- linkedin.com/in/muhammad-ali-waris-khan-97a8ba146
- mawk-portfolio.vercel.app